

Jiawei Lian, Curriculum Vitae

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🌐 [Research Gate](#)

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-**Birth Date:** 01/1997

-**Status:** Final-year PhD student, PolyU and NWPU joint training leading to dual PhD

-**Research Interest:** AI Safety, Large language model, Computer vision, Remote sensing

-**Postal Address:** Room 505, Building CF, The Hong Kong Polytechnic University, Hung Hom, Kowloon, Hong Kong

Education

- 01/2024 – Present 📖 Ph.D., Artificial Intelligence and Signal Processing, Advisor: [Lap-Pui Chau](#)
The Hong Kong Polytechnic University (PolyU)
Thesis: *Towards Exploring the Adversarial Vulnerability of Deep Learning Models*
- 04/2022 – Present 📖 Ph.D., Information and Communication Engineering, Advisor: [Shaohui Mei](#)
Northwestern Polytechnical University (NWPU)
Thesis: *Towards Exploring the Adversarial Vulnerability of Deep Learning Models*
- 09/2019 – 03/2022 📖 M.E., Control Engineering, Advisor: [Junhong He](#)
Northwestern Polytechnical University
Thesis: *Research on Rapid Detection of Surface Defects Based on Deep Learning*
- 09/2015 – 07/2019 📖 B.E., Automation, Advisor: Junlin Zhu
JiangXi University of Science and Technology
Thesis: *Design of 4R Joint Robot Control System based on PROFIBUS and T-CPU*

Research Publications

Journal Articles

1. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Xiaofei Wang, Yingjie Lu, Shaohui Mei, Lap-Pui Chau. "Revealing the Intrinsic Ethical Vulnerability of Aligned Large Language Models." *Nature Communications (NC)*, 2026. **(Accepted)**
2. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. "PADetBench: Towards Benchmarking Physical Attacks against Object Detection." *Knowledge-Based Systems (KBS)*, 2025.
3. Shaohui Mei, **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Lap-Pui Chau. "A Comprehensive Study on the Robustness of Image Classification and Object Detection in Remote Sensing: Surveying and Benchmarking." *Journal of Remote Sensing (JRS)*, 2024. *(The first author is the PhD advisor)*
4. **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. "CBA: Contextual Background Attack against Optical Aerial Detection in the Physical World." *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 61, pp. 1–16, 2023.
5. **Jiawei Lian**, Shaohui Mei, Shun Zhang, Mingyang Ma. "Benchmarking Adversarial Patch against Aerial Detection." *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, vol. 60, pp. 1–16, 2022.
6. **Jiawei Lian**, Junhong He, Yun Niu, Tianze Wang. "Fast and Accurate Detection of Surface Defect Based on Improved YOLOv4." *Assembly Automation (AA)*, vol. 42, no. 1, pp. 134–146, 2021.
7. **Jiawei Lian**, Junhong He, Yun Niu, Tianze Wang. "Rapid Detection of Surface Defects Based on Multi-scale Compression CNN." *Computer Integrated Manufacturing System (CIMS)*, vol. 28, no. 11, pp. 3624, 2021.

Conference Papers

1. **Jiawei Lian**, Jianhong Pan, Lefan Wang, Yi Wang, Shaohui Mei, Lap-Pui Chau. "Semantic Representation Attack against Aligned Large Language Models." *Advances in Neural Information Processing Systems (NeurIPS)*, 2025.
2. **Jiawei Lian**, Xiaofei Wang, Yuru Su, Mingyang Ma, Shaohui Mei. "Contextual Adversarial Attack against Aerial Detection in the Physical World." *IEEE International Geoscience and Remote Sensing Symposium (IGARSS)*, pp. 6632–6635, 2023.

Preprints

1. **Jiawei Lian**, Shaohui Mei, Xiaofei Wang, Yi Wang, Lefan Wang, Yingjie Lu, Mingyang Ma, Lap-Pui Chau. "Attack Anything: Blind DNNs via Universal Background Adversarial Attack." *arXiv preprint arXiv:2409.00029*, 2024. (**Under review, Neural Networks**)

China Patents

- | | | |
|------------|---|--|
| 16/05/2023 | ■ | A method and device for generating adversarial camouflage patterns against object detection. (Patent acceptance number: 202310547602.8) |
| 16/12/2022 | ■ | A method and device for physical attack-oriented self-adaptive adversarial patch generation. (Patent acceptance number: 202211617496.8) |
| 15/12/2022 | ■ | A method and device for generating adversarial patches for unobstructed physical adversarial attacks. (Patent acceptance number: 202211612346.8) |
| 05/06/2021 | ■ | A fast surface defect detection method based on the lightweight convolutional neural network. (Patent acceptance number: 202110627893.2) |

Miscellaneous Experience

Honours and Awards

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| 2023-2025 | ■ | Outstanding Graduate Student |
| 2023 | ■ | Suzhou "Talent Training" Scholarship |
| 2023-2024 | ■ | First Class Academic Scholarship |

Involved Projects

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| 03/2024-Present | ■ | Research on exploring the adversarial vulnerability of large language models. |
| 09/2022-02/2024 | ■ | Research on the robustness of visual perception based on deep learning models. |
| 12/2021-08/2022 | ■ | Research on background adversarial perturbations for physical attack. Relevant research achievements received an interview invitation from the British authoritative magazine New Scientist. |

Certifications

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| 04/2022 | ■ | IELTS: Overall band score 7.0 (Listening 7.5, Reading 7.5, Writing 6.5, Speaking 6.0) |
| 07/2017 | ■ | Fitness Coach Licentiate (FCL): Awarded by The Ministry of Human Resource and Social Security, P.R. China. |

Internship

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| 07/2021-09/2021 | ■ | Internship at AVIC Xi'an Flight Automatic Control Research Institute. |
| 07/2020-09/2020 | ■ | Internship at Xi'an Institute of Modern Control Technology. |